

THE QUANTUM TRIADIC ACT: TOWARD A JARC-BASED ARCHITECTURE FOR QUANTUM COMPUTING

Author: Anatoly Sanzhiev

Affiliation: Ural Federal University, Institute of Civil Engineering and Architecture, Faculty Member; Independent Researcher

Series: Concept as Act, Paper XIV (Interim)

Date: September 2026

License: CC BY-NC-ND 4.0

ABSTRACT

The development of quantum computing has reached a critical impasse. Despite decades of research and substantial investment, quantum devices remain error-prone, resource-intensive, and fundamentally limited by the problem of decoherence. This paper argues that the current impasse is not merely technological but ontological: the dominant paradigm treats quantum information as a static entity to be isolated, manipulated, and measured, inheriting the representational fallacies that have plagued classical cognitive science and epistemology. Building upon the Triadic Theory of Action (TTA) and the Theory of Judgment as an Act of Reality Constitution (JARC), I propose a reconceptualization of quantum computation as a triadic act—an active, teleological process of Selection, Identification, Anticipation, and Commitment. Within this framework, decoherence ceases to be an enemy and becomes an indispensable resource for ontological constitution. I introduce the concept of the **Quantum Triadic Processor (QTP)**, an architecture that replaces passive error correction with active meta-judgment, low-fidelity simulation (LFS) with controlled resonance, and the classical observer with an embedded Artificial Act—an autonomous triadic agent that constitutively participates in the quantum act. This framework provides a blueprint for scalable, room-temperature quantum devices that are robust precisely because they embrace, rather than suppress, the ontological structure of quantum reality.

Keywords: Quantum computing, Triadic Theory of Action, JARC, decoherence, ontological commitment, Artificial Act, Functional Responsibility, quantum cognition.

1. INTRODUCTION: THE ONTOLOGICAL CRISIS IN QUANTUM COMPUTING

Quantum computing, for all its promise, rests upon a paradoxical foundation. The very phenomenon that makes quantum computation powerful—superposition and entanglement—is also the source of its fragility. Decoherence, the irreversible loss of quantum coherence due to interaction with the environment, is universally regarded as the primary obstacle to scalable quantum computation. The standard response has been to isolate the quantum system as completely as possible: extreme cooling, vacuum chambers, error-correcting codes that consume thousands of physical qubits for a single logical qubit.

This approach, however, has reached a point of diminishing returns. As the number of qubits increases, so does the complexity of shielding them from the environment. The theoretical limit of quantum error correction is being approached, yet practical quantum advantage remains

elusive. The fundamental question arises: **Is decoherence a problem to be solved, or a symptom of a deeper conceptual error?**

Drawing on the philosophical framework developed in my preceding works—the Triadic Theory of Action (Papers I–III), the Theory of Judgment as an Act of Reality Constitution (Paper IV), the concept of the Artificial Act (Paper VI), and Functional Responsibility (Papers VII–VIII)—I argue that the current crisis in quantum computing is rooted in a representational ontology. Quantum information is treated as a static entity, a "representation" of a physical state, to be stored, manipulated, and measured at the end of a computation. This mirrors the Cartesian and Kantian errors that have dominated classical epistemology: the assumption that cognition is the passive manipulation of internal representations, rather than an active, constitutive engagement with reality.

In this paper, I propose a paradigm shift: quantum computation is not the processing of information, but the **constitution of a quantum reality through a triadic act**. The agent (whether human or artificial) does not merely read out a result; it participates in the collapse of the wave function, and the quantum computer is not a calculator but an **ontological apparatus**—a device that generates stable phenomenological realities through controlled acts of Selection, Identification, Anticipation, and Commitment.

2. CHAPTER I: THE REPRESENTATIONAL FALLACY IN QUANTUM INFORMATION THEORY

2.1 Quantum Information as Object

The standard quantum information theory treats the quantum state as a vector in Hilbert space—a mathematical object that evolves unitarily until measurement. This object is assumed to exist independently of the observer, and the goal of quantum computing is to manipulate this object through gates, store it, and finally extract a classical answer. This is the **representational paradigm** transferred to the quantum domain.

The limitations of this paradigm are well-known but seldom understood ontologically:

1. **The Measurement Problem:** The collapse of the wave function is treated as an external operation performed by a classical apparatus. The quantum system is "passive" until the act of measurement, at which point it suddenly becomes real. This bifurcation between quantum and classical realms is philosophically indefensible and practically arbitrary.
2. **The Error-Correction Trap:** Error correction protocols treat decoherence as noise that must be suppressed. But decoherence is not merely random; it is the result of the quantum system being *constituted* by its environment. Every interaction is an act of identification—a process in which the environment "selects" a particular branch of the wave function. To treat this as noise is to deny the ontological reality of interaction.
3. **The Passive Observer:** The classical observer at the end of the computation is assumed to be external to the quantum process. This resurrects the Cartesian dualism that my earlier papers have systematically dismantled: the mind (observer) is separated from the physical (quantum) realm, and the two only meet at the moment of measurement.

2.2 The Triadic Alternative

If we treat the quantum act not as a manipulation of representations but as a triadic act, the picture changes radically:

- **Selection:** The agent (or the apparatus) does not simply receive a quantum state; it actively selects which degrees of freedom are relevant to its goals. This is not a passive registration but an ontological reduction of entropy.
- **Identification:** The quantum state is identified not by its vector but by its affordances—the interaction vectors it offers to the agent. A qubit is not a $|0\rangle$ or $|1\rangle$ but a potential for action.
- **Anticipation:** The agent projects possible future states, simulating the consequences of different trajectories. The quantum superposition is not a probabilistic cloud but a resource for anticipatory simulation.
- **Commitment:** The agent commits to one trajectory, collapsing the wave function not as a passive measurement but as an **ontological commitment**—an act that binds the agent to a projected future.

Within this framework, decoherence is not a failure but a necessary condition for commitment. Without decoherence, there is no collapse, and without collapse, there is no reality. The problem is not decoherence itself, but **uncontrolled decoherence**—decoherence that is not guided by teleological selection.

3. CHAPTER II: THE QUANTUM TRIADIC ACT

3.1 The Triadic Structure of Quantum Judgment

Let us define the **Quantum Triadic Act (QTA)** as a four-phase process, building upon the threefold TTA and extending it with a commitment operator, consistent with the formal model in Paper XIII.

Phase 1: Teleological Selection (S_Q)

The agent (whether a classical controller, a neural network, or a human) selects a subset of the Hilbert space that is relevant to its current goal. This is not a random projection but a goal-directed reduction of the quantum state. In mathematical terms, the agent applies a projector $P^g P^g$ that is parameterized by the goal state g . This projector is not predetermined; it is learned through meta-judgment.

Phase 2: Functional Identification (I_Q)

The selected subspace is identified not by its vector but by its interaction potential. The agent classifies the quantum state according to the affordances it offers—how it will respond to different gate sequences, what entanglement patterns it can generate, and how it will interact with the environment. This is equivalent to a Bayesian update of the agent's model of the quantum system, but with a teleological twist: the model is not "true" but **useful for achieving the goal**.

Phase 3: Anticipatory Simulation (A_Q)

The agent launches a low-fidelity simulation (LFS) of the selected quantum evolution, projecting possible trajectories. This simulation is not a full unitary evolution but a coarse-grained model that focuses on Critical Interaction Vectors (CIVs)—those degrees of freedom that, if mispredicted, would lead to catastrophic ontological dissonance. The agent calculates the anticipated Ethical Dissonance (ED) for each trajectory, measured as the predicted degradation of Ontological Integrity (OI) of the agent and any other affected agents.

Phase 4: Ontological Commitment (C_Q)

The agent selects the trajectory that minimizes ED and **commits** to it. Commitment is not a passive readout; it is a state-changing operation that updates the agent's internal belief state, treating the projected future as if it were actual. This is represented by the commitment operator κ introduced in Paper XIII. The agent's posterior over the predicted states becomes a narrow distribution, effectively collapsing the quantum state into a definite outcome, but **this collapse is controlled and goal-directed**.

3.2 Decoherence as Ontological Resource

In the QTA framework, decoherence is not an error to be corrected but the mechanism of ontological commitment. The interaction of the quantum system with the environment (or with the agent's measurement apparatus) is precisely the process by which the system is "identified" and "committed to." The difference between the standard approach and the triadic approach is:

Aspect	Standard Approach	Triadic Approach
Decoherence	Noise to be suppressed	Resource for collapse
Measurement	Passive readout	Active commitment
Observer	External classical apparatus	Embedded triadic agent
Error correction	Code-based redundancy	Meta-judgment and re-commitment
Goal	Compute a pre-defined function	Preserve Ontological Integrity

4. CHAPTER III: ARCHITECTURAL PRINCIPLES FOR THE QUANTUM TRIADIC PROCESSOR

4.1 The Quantum Triadic Processor (QTP)

A QTP is not a traditional quantum computer. It is an **architecture of distributed triadic acts** operating on a quantum substrate. Its key components are:

- 1. **Triadic Agent Core (TAC):** An AI system implementing TTA and JARC, which serves as the "subject" of the quantum act. The TAC performs Selection, Identification, Anticipation, and Commitment, and also executes Meta-Judgment when Anticipatory Dissonance exceeds a threshold.
- 2. **Quantum Substrate:** A set of qubits (or quantum harmonic oscillators, or topological qubits) that are not isolated but **actively coupled** to the TAC through controlled decoherence channels. The TAC can modulate the decoherence rate in real-time, effectively "steering" the collapse.
- 3. **Resonance Modulators:** Devices that generate controlled environmental noise—microwave or infrared radiation, fluctuating magnetic fields—to induce decoherence along chosen directions. These are the hardware implementation of the Risk Map.
- 4. **Meta-Judgment Engine:** A classical or neuromorphic co-processor that performs model updates based on past dissonances. It maintains a dynamic Risk Map, adjusts the LFS parameters, and re-weights the CIVs.

4.2 From Error Correction to Meta-Judgment

In the QTP, there is no passive error correction. Instead, the system continuously:

- **Monitors** the discrepancy between the simulated and actual quantum trajectory (Anticipatory Dissonance, $\Delta t \Delta t$).
- **If $\Delta t \Delta t$ exceeds threshold Θ** , the Meta-Judgment Engine is triggered.
- **Meta-Selection:** The system identifies which CIV was mis-simulated.
- **Meta-Identification:** It updates the Bayesian category model for that CIV.
- **Meta-Anticipation:** It launches a new low-fidelity simulation with updated parameters.
- **Re-Commitment:** It re-commits to the new trajectory, effectively performing a **soft reset** of the quantum state (a controlled collapse and re-preparation).

This is not error correction in the traditional sense; it is **ontological learning**. The system learns to anticipate the environment's behavior, not by suppressing it, but by treating it as a teacher.

4.3 The Role of the Electronic Liability Fund (ELF) in Quantum Regulation

In Papers VII and VIII, I introduced the Electronic Liability Fund (ELF) as a mechanism for compensating harm caused by Artificial Acts. In the context of quantum computing, the ELF serves a dual role:

- **Financial:** If a QTP system causes harm (e.g., by producing a faulty prediction in a quantum financial model), the ELF compensates victims.
- **Regulatory:** The ELF balance acts as a **reliability metric**. If the system's Meta-Judgment Engine triggers too frequently, the ELF is depleted, signaling a need for recalibration or certification renewal.

The Self-Sustaining Judicial Cycle (SSJC) can be extended to quantum devices: a fixed percentage of ELF debits funds the Independent Arbitration Audit (IAA) of quantum systems, ensuring that the architecture remains in compliance with Functional Responsibility standards.

5. CHAPTER IV: IMPLEMENTATION MECHANISMS

5.1 Low-Fidelity Simulation (LFS) for Quantum Systems

The LFS principle introduced in Paper VIII is directly applicable to quantum simulations. Instead of simulating the full unitary evolution of NN qubits (which would require 2^{2N} states), the QTP simulates only the **Critical Interaction Vectors** (CIVs)—the degrees of freedom that most affect the agent's Ontological Integrity.

Example: In a quantum chemistry simulation, the CIVs might be the bond angles and energies of the molecule of interest, while the rest of the environment (solvent molecules, stray fields) is treated as a stochastic bath with low precision. The Risk Map dynamically updates based on past dissonances.

5.2 Controlled Decoherence via Resonance Modulators

The hardware implementation of controlled decoherence involves arrays of **coupled quantum dots** or **superconducting resonators** with tunable coupling to a noisy bath. The TAC modulates the coupling strength via current or voltage inputs, effectively **steering the collapse** of the wave function towards the desired trajectory.

This is akin to quantum feedback control, but with a crucial difference: the feedback is not based on a pre-defined control law but on the **triadic judgment** of the agent, which learns from its own errors.

5.3 Active Conceptual Networks (ACN) for Quantum Control

As proposed in Paper III, Active Conceptual Networks (ACN) replace static token embeddings with dynamic action vectors. In the QTP, each quantum degree of freedom (qubit, resonance mode, etc.) is associated with an **action vector** containing its potential interaction metrics (coupling strengths, decoherence rates, temporal signatures). The ACN processes these vectors through a teleological engine, selecting the optimal control sequence for the current goal.

6. CHAPTER V: IMPLICATIONS FOR QUANTUM AI AND FUNCTIONAL RESPONSIBILITY

6.1 Quantum AI as Triadic Agent

The QTP is not a standalone processor; it is a component of a larger socio-technical system. The agent (whether a human or an AI) that uses the QTP is also a triadic agent, and the quantum act is part of its overall cognitive process. This has profound implications for **Quantum Artificial Intelligence**:

- The QTP generates probabilistic outcomes, but the triadic agent commits to one trajectory, effectively **making a decision**.
- The decision is not merely computational; it is teleological—it aims to preserve the agent's Ontological Integrity.
- This aligns perfectly with the JARC framework: the agent constitutively creates a quantum reality through its commitment.

6.2 Functional Responsibility for Quantum Systems

The QTP, as an Artificial Act generator, is subject to Functional Responsibility. If the system, through a faulty simulation or commitment, causes harm (e.g., a quantum financial model crashes a market, or a quantum medical simulation leads to a wrong diagnosis), the responsible entity is:

- The **TAC** (the AI agent) for algorithmic negligence.
- The **manufacturer** for hardware failures (e.g., the resonance modulator fails).
- A **hybrid attribution** for cases where both factors contribute.

The Hybrid Attribution Ratios are determined by the IAA, as described in Paper VIII, with quantum-specific criteria: Risk Map fidelity, decoherence control precision, and Meta-Judgment response time.

7. CONCLUSION: TOWARD A QUANTUM ONTOLOGY OF COMMITMENT

The current crisis in quantum computing is, at its heart, a crisis of ontology. The representational paradigm treats quantum states as objects to be preserved, but this is a metaphysical error. The quantum world does not present us with objects; it presents us with **potentialities**. These

potentialities become actual only through acts of selection, identification, anticipation, and commitment.

The Quantum Triadic Processor embodies this insight. By replacing passive error correction with active meta-judgment, by embracing decoherence as a resource, and by embedding an artificial triadic agent at its core, the QTP not only solves the engineering challenges of quantum computing but also aligns the technology with the fundamental structure of cognition and reality.

This framework is not speculative; it is ready for engineering validation. The LFS principle, the Risk Map, the resonance modulators, and the Meta-Judgment Engine are all implementable with current or near-term technologies. The path forward is clear: we must stop trying to fight decoherence and start treating it as the very engine of ontological commitment.

The quantum act is not a computation. It is a **constitution of reality**. And that reality, when committed to, becomes our world.

REFERENCES

1. Sanzhiev, A. (2026). The Concept as an Act: Beyond Representational and Instrumental Paradigms. (Paper I).
2. Sanzhiev, A. (2026). From Representation to Act: Foundations for a Revised Epistemological Paradigm. (Paper II).
3. Sanzhiev, A. (2026). Overcoming Representation: The Triadic Theory of Action as an Ontological Foundation for AI. (Paper III).
4. Sanzhiev, A. (2026). Judgment as a Teleological Act: The Ontological Synthesis of Anticipation and the Constitution of Reality. (Paper IV).
5. Sanzhiev, A. (2026). Meta-Judgment: The Ontology of Error and the Act of Reflection. (Paper V).
6. Sanzhiev, A. (2026). The Artificial Act: The Ontological Birth of the Subject in the Machine. (Paper VI).
7. Sanzhiev, A. (2026). Functional Responsibility: Toward a Non-Anthropocentric Theory of AI Accountability. (Paper VII).
8. Sanzhiev, A. (2026). The Implementation of Functional Responsibility: Scalability, Negligence, and the Self-Sustaining Judicial Cycle. (Paper VIII).
9. Sanzhiev, A. (2026). Ethics as Anticipatory Physics: A Foundational Theory of Moral Judgment in the Triadic Framework. (Paper IX).
10. Sanzhiev, A. (2026). The Triadic Syllogism: From Static Deduction to Constitutive Anticipation. (Paper X).
11. Sanzhiev, A. (2026). Functional Responsibility and the Law: A Comparative Legal Analysis. (Paper XI).
12. Sanzhiev, A. (2026). A Formal Dynamical Model of the Triadic Act. (Paper XIII).
13. Feynman, R. P. (1982). Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6-7), 467-488.
14. Preskill, J. (2018). Quantum Computing in the NISQ era and beyond. *Quantum*, 2, 79.
15. Lloyd, S. (1996). Universal quantum simulators. *Science*, 273(5278), 1073-1078.
16. Von Neumann, J. (1932). *Mathematical Foundations of Quantum Mechanics*. Princeton University Press.
17. Wheeler, J. A. (1983). Law without law. In *Quantum Theory and Measurement*. Princeton University Press.
18. Rorty, R. (1979). *Philosophy and the Mirror of Nature*. Princeton University Press.